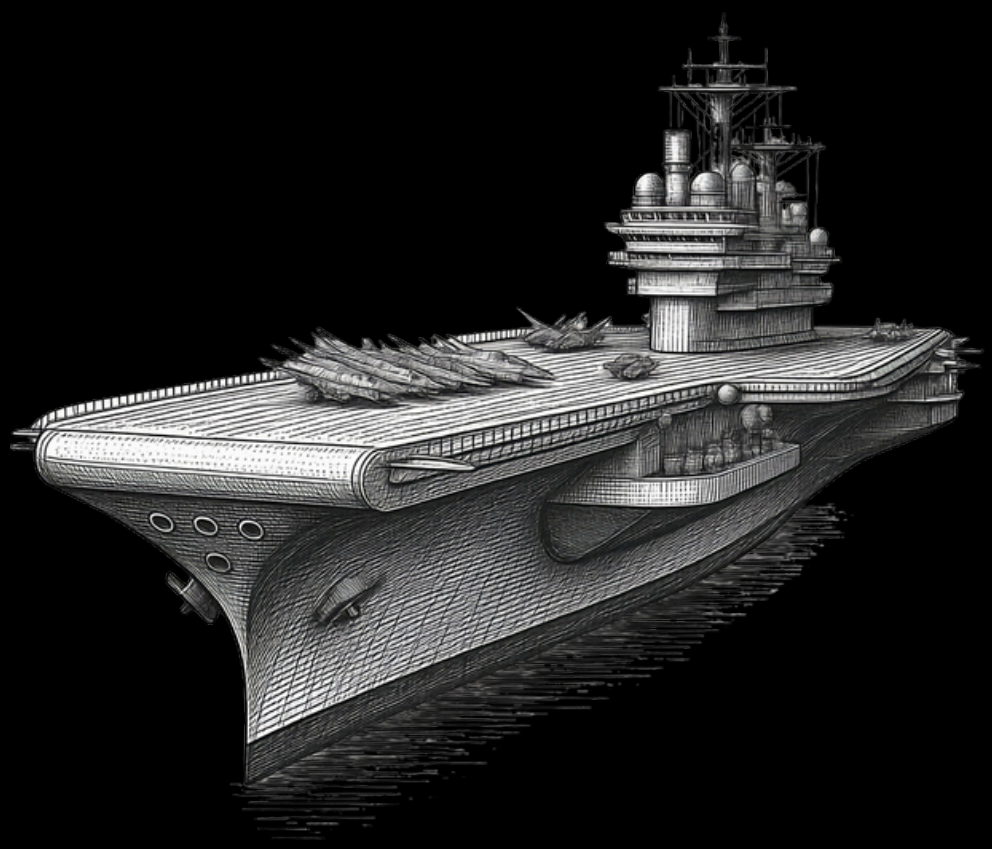
The Dual-Use Problem

When Commercial Cloud Becomes a Military Asset



An Analysis of How the Pentagon's JWCC and JADC2 Frameworks Have Permanently Collapsed the Boundary Between Civilian and Military Compute Infrastructure

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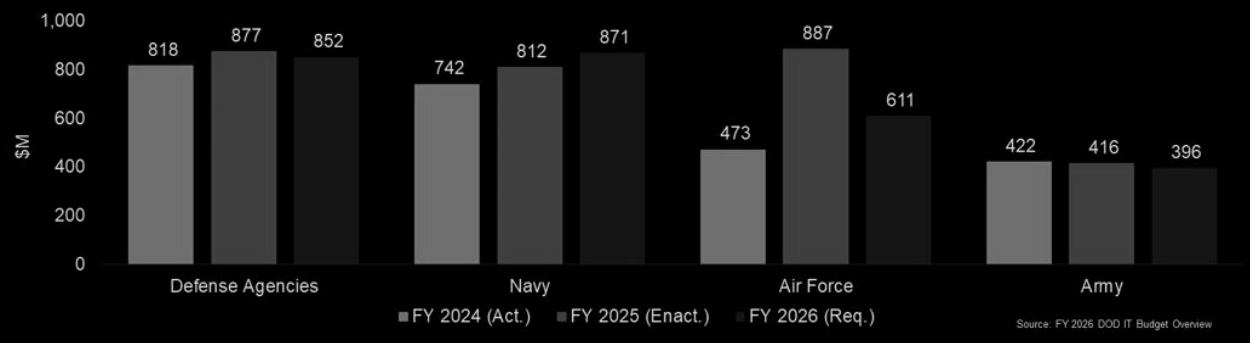
I. The Collapse of the Civil-Military Boundary

The distinction between civilian cloud infrastructure and military compute capability has effectively ceased to exist. The Pentagon's Joint Warfighting Cloud Capability contract, its Joint All-Domain Command and Control architecture, and the broader integration of commercial AI into defense operations have created a world in which every major hyperscaler data center is, by design or by proximity, a dual-use asset.

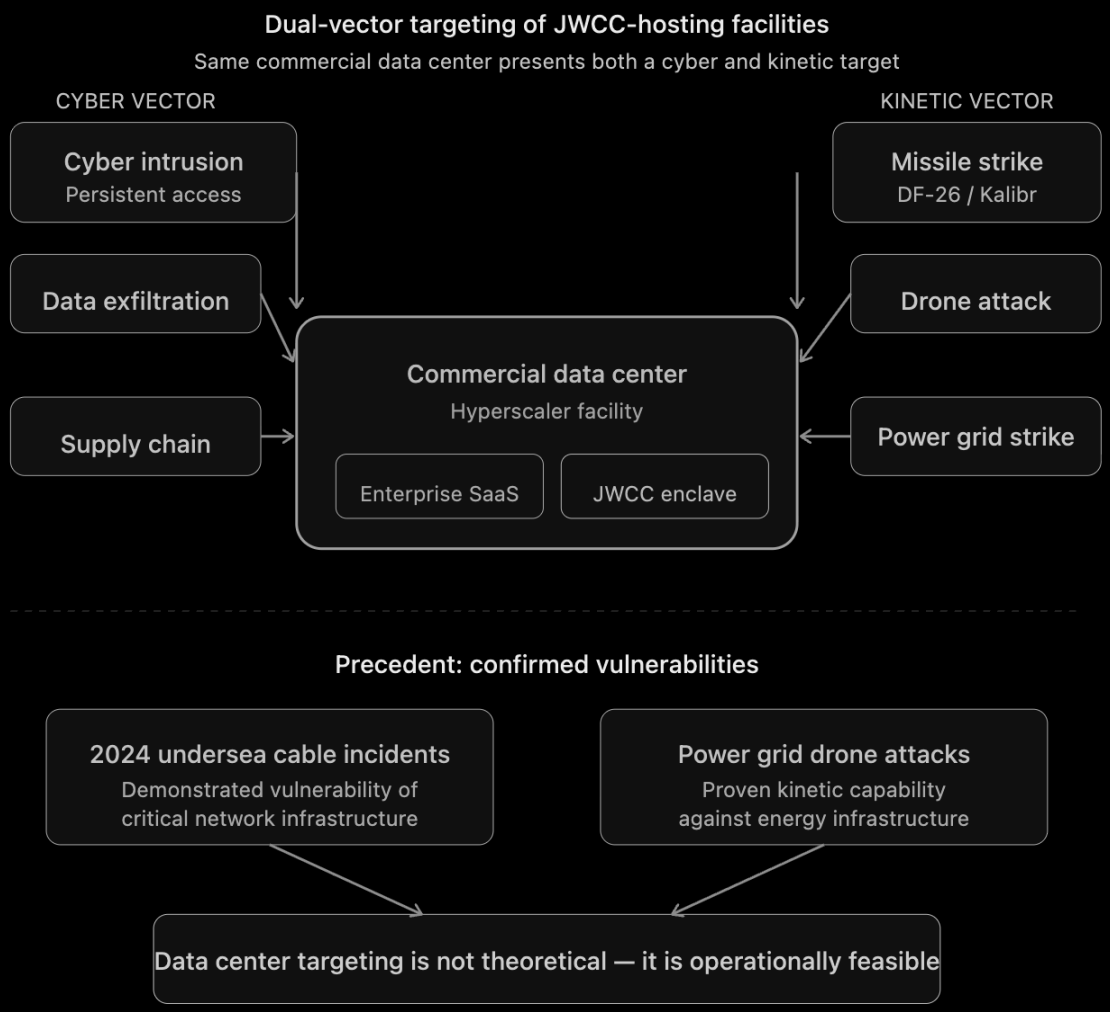
- **JWCC formalized the entanglement.** The Department of Defense's $9 billion Joint Warfighting Cloud Capability contract, awarded in December 2022 to Amazon Web Services, Microsoft Azure, Google Cloud, and Oracle, replaced the failed single-vendor JEDI approach with a multi-cloud architecture. JWCC workloads run on the same physical infrastructure that hosts commercial customers. Classification boundaries are enforced through logical separation, software partitions, not physically distinct facilities. The hyperscaler data centers serving Fortune 500 enterprises are simultaneously processing classified and unclassified DoD workloads.
- **JADC2 extends military dependency on commercial compute to the tactical edge.** The Joint All-Domain Command and Control framework envisions real-time sensor-to-shooter data fusion across all military domains. Achieving the latency and throughput this demands requires distributed compute at a scale only commercial cloud providers can deliver. The Air Force's ABMS, the Army's Project Convergence, and the Navy's Project Overmatch all depend on commercial cloud backends.
- **The AI acceleration made separation impossible.** The Pentagon's adoption of commercial AI tools, from Palantir's Maven Smart System to Microsoft's classified GPT-4 deployment on the CIA's C2E cloud, means the same foundation models, GPU clusters, and inference pipelines serving civilian applications are being fine-tuned for intelligence analysis, targeting, and operational planning. The infrastructure is not adjacent. It is identical.
- **The numbers confirm systemic integration.** The Department of Defense obligated over $7.3 billion on cloud services between FY2020 and FY2024, with spending accelerating annually. The intelligence community's Commercial Cloud Enterprise contract with AWS, valued at up to $10 billion, hosts data up to Top Secret/SCI on commercially operated infrastructure. The U.S. national security establishment is now one of the largest customers of hyperscaler cloud services.



II. Targeting Doctrine: The Infrastructure Becomes the Objective

The dual-use reality has immediate consequences for how adversaries conceptualize targeting. In any near-peer conflict scenario, commercial data centers supporting military workloads become legitimate military objectives under the law of armed conflict, and rational strategic targets regardless of legal classification.

- **International humanitarian law permits targeting dual-use infrastructure.** Additional Protocol I to the Geneva Conventions defines military objectives as objects which by their nature, location, purpose, or use make an effective contribution to military action. A data center processing JWCC workloads satisfies this definition. The principle of distinction becomes operationally incoherent when the same servers process a retail company's inventory system and a combatant command's operational planning data.
- **Adversary doctrine already accounts for this vulnerability.** Chinese military strategy, articulated through systems destruction warfare, explicitly targets the information systems enabling adversary decision-making. Russian doctrine similarly emphasizes strikes against critical information infrastructure in the initial period of war. Both frameworks logically extend to commercial data centers hosting military compute, the doctrinal question is not whether to strike them but when and in what sequence.
- **Cyber and kinetic targeting converge on the same facilities.** A commercial data center hosting JWCC workloads presents both a cyber and kinetic target. Adversaries have persistent incentives to compromise hyperscaler environments containing classified enclaves. In kinetic conflict, the same facilities become candidates for physical destruction. The 2024 undersea cable incidents and documented vulnerability of power infrastructure to drone attack confirm that data center targeting is not theoretical.



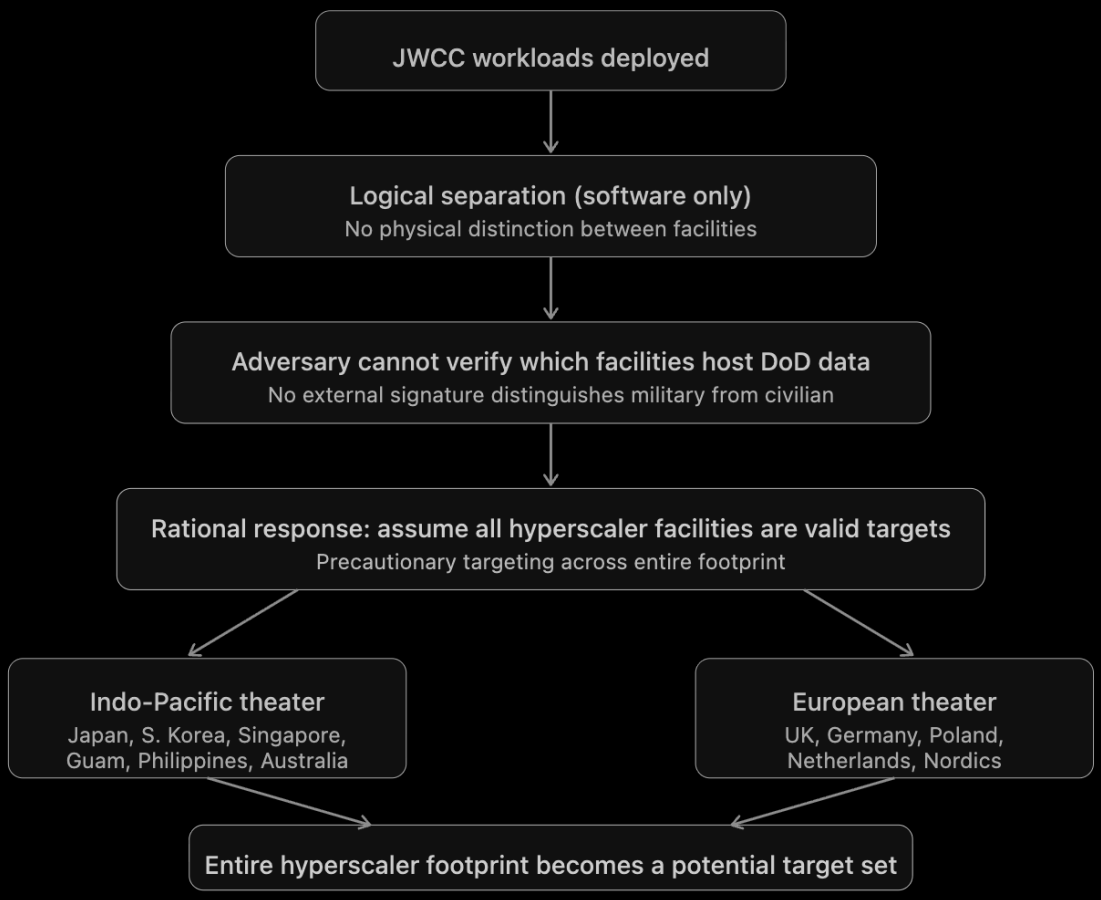
III. Geopolitical Spillover: Why Every Host Nation Bears the Risk

The dual-use problem is not confined to the continental USA. U.S. hyperscalers operate data centers in 30 countries, and JWCC's architecture explicitly contemplates deployed and edge cloud capabilities in allied/partner nations. This globalizes the targeting risk.

- **Allied hosting creates shared vulnerability.** AWS, Microsoft, and Google operate major cloud regions in Japan, South Korea, Germany, the United Kingdom, Australia, and across NATO. If these facilities support, or could plausibly support, U.S. military workloads, they become potential targets in a near-peer conflict. Host nations inherit a targeting risk they may not have explicitly consented to and may not fully understand.
- **The plausible deniability problem compounds the risk.** Because JWCC workloads are logically rather than physically separated, an adversary cannot verify whether a specific data center processes military data. The rational response is to assume any hyperscaler facility in a strategically relevant geography could support military operations. This transforms the entire hyperscaler footprint in the Indo-Pacific and European theaters into a potential target set.
- **Emerging markets face asymmetric exposure.** As hyperscalers expand into Southeast Asia, the Middle East, and Africa, they bring the dual-use problem with them. A data center in the Philippines or Bahrain serving local enterprise customers may simultaneously function as a forward node for military AI inference. Host governments acquire infrastructure investment and connectivity, but also acquire an adversary's reason to target facilities on their soil.

The plausible deniability problem

Why logical separation expands the target set rather than protecting it



IV. Insurance, Liability, and the Pricing of Dual-Use Risk

The financial infrastructure surrounding data centers, insurance, sovereign guarantees, lending covenants, and corporate risk frameworks, has not adapted to the dual-use reality. This represents a material underpriced risk across the data center investment landscape.

1. **War exclusion clauses create coverage gaps.** Standard property and casualty insurance policies contain war and terrorism exclusions. If a commercial data center is struck because it hosts military workloads, insurers will contest coverage.
2. **Sovereign hosts face unquantified contingent liabilities.** Governments courting hyperscaler investment through tax incentives and subsidized power may inadvertently accept contingent military risk. If a data center on sovereign territory is targeted in a conflict the host nation is not party to, the consequences extend far beyond the facility itself.
3. **Credit markets have not priced the risk.** Data center REITs, infrastructure bonds, and project finance vehicles assume peacetime operating continuity. The dual-use problem introduces tail risk, low probability, catastrophic impact, not reflected in credit spreads, cap rates, or insurance premiums. As awareness grows, repricing is likely, particularly for assets in contested geographies.

V. Strategic Implications for Sovereign Infrastructure

The dual-use problem strengthens the case for sovereign-controlled digital infrastructure that is architecturally distinct from hyperscaler environments, and reframes data localization not merely as a regulatory preference but as a national security imperative.

- **Sovereign cloud reduces targeting exposure.** A nationally owned data center that does not host foreign military workloads is not a legitimate military objective under IHL, absent its own military function. Governments seeking to insulate digital infrastructure from great-power conflict dynamics have a structural incentive to build sovereign capacity rather than rely exclusively on hyperscaler tenancy.
- **Physical separation is the only credible guarantee.** Logical separation, the JWCC model, may satisfy peacetime compliance, but it does not satisfy an adversary's targeting calculus. Only physical separation of civilian and military compute into distinct facilities with distinct ownership provides a credible basis for claiming protected status under the law of armed conflict.
- **Non-aligned nations have the most to gain from architectural independence.** Countries in the Global South face a strategic choice: accept hyperscaler tenancy with its dual-use risk, or invest in sovereign infrastructure that preserves neutrality. For nations unwilling to be drawn into great-power conflict through the infrastructure choices of their technology vendors, sovereign data center capacity is a prerequisite for genuine non-alignment.
- **The investment case for sovereign infrastructure expands.** The dual-use dynamic adds a national security dimension to the existing economic and regulatory case for sovereign data centers. DFIs, sovereign wealth funds, and infrastructure investors should assess target markets for geopolitical exposure to hyperscaler dual-use risk alongside localization mandates and commercial demand. Markets with high hyperscaler dependency near contested theaters represent the highest-priority opportunities.

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VI. Conclusion

The boundary between civilian and military compute infrastructure is not blurring. It has already collapsed. The Pentagon's JWCC and JADC2 architectures, combined with the wholesale integration of commercial AI into defense operations, have made every major hyperscaler data center a potential node in the American military's information architecture, and therefore a potential target for its adversaries.

This is not a future risk. It is a present condition with immediate implications for infrastructure investors, host-nation governments, corporate risk managers, and the international legal frameworks governing armed conflict. The data center industry's operating assumption, that commercial infrastructure exists in a domain separate from military operations, is no longer tenable. Sovereign-aligned, architecturally independent digital infrastructure is the rational response for any nation that takes seriously the protection of its critical systems and the neutrality of its territory.

Analysis draws on primary review of Department of Defense contract awards, published doctrine and strategy documents, congressional testimony, and publicly available reporting on JWCC, JADC2, and related programs. Legal analysis references Additional Protocol I to the Geneva Conventions, the DoD Law of War Manual, and relevant international humanitarian law scholarship. Geopolitical risk assessments incorporate open-source intelligence on adversary military doctrine and targeting frameworks. Infrastructure data draws on JLL, Cushman & Wakefield, Structure Research, and DC Byte. All assessments as of March 2026.

The Tatari Institute is the research division of Tatari, producing institutional-grade analysis at the intersection of AI infrastructure, sovereign digital policy, and emerging-market investment. Our research serves sovereign wealth funds, development finance institutions, and infrastructure investors navigating the global build-out of digital infrastructure.

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